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Leading Through Complexity: An Integrative Review and Contingent Innovation Framework

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Short Bio

I am a Visiting Professor at the University of Nicosia's School of Business (AACSB-accredited) and a member of the Knowledge Management, Innovation and Strategy Center (KISC). My research develops and applies evolutionary, systems-oriented frameworks (macro meso–micro) to explain how firms, ecosystems, and institutions co-evolve under global restructuring—linking international political economy, business ecosystems, entrepreneurship, and sectoral systems (including energy). I have authored 100+ scholarly outputs, including peer-reviewed journal articles, chapters in edited volumes, and conference papers.

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Data Availability Statement (DAS)

The entire dataset is included in Supplementary File S1 and Supplementary File S2.

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Abstract

Purpose: *This integrative review synthesizes research on adaptive, agile, complexity and ambidextrous leadership and innovation under VUCA (volatility, uncertainty, complexity and ambiguity) to clarify how leaders orchestrate context-sensitive ("contingent") innovation.*

Design/methodology/approach: *Guided by Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020, we searched Scopus (29 December 2025), screened 85 records and synthesized 34 journal articles (2015–2025). We appraised evidence (Mixed Methods Appraisal Tool (MMAT)-informed for empirical papers; conceptual-rigor checklist for conceptual papers) and analyzed it using a concept-centric coding matrix.*

Findings: *Four themes recur: (1) leadership as sensemaking/framing and enabling experimentation; (2) innovation as exploration–exploitation balancing (ambidexterity), often expressed at the level of an evolving innovation initiative portfolio; (3) psychological safety, trust, organizational support and self-efficacy as mechanisms for innovative work behavior; and (4) dynamic capabilities that transform experiments into reconfiguration. We develop a four-proposition framework that integrates these themes by specifying mediation pathways and boundary conditions (portfolio governance routines and coordination overload).*

Originality: *We define contingent innovation capability as an organizational orchestration capability—comprising sensing, portfolio governance and integration/reconfiguration routines—and explain how adaptive leadership behaviors develop and activate it to couple safe-to-fail experimentation with disciplined integration as contingencies shift.*

Research limitations/implications: *Theory-building synthesis; propositions require empirical testing.*

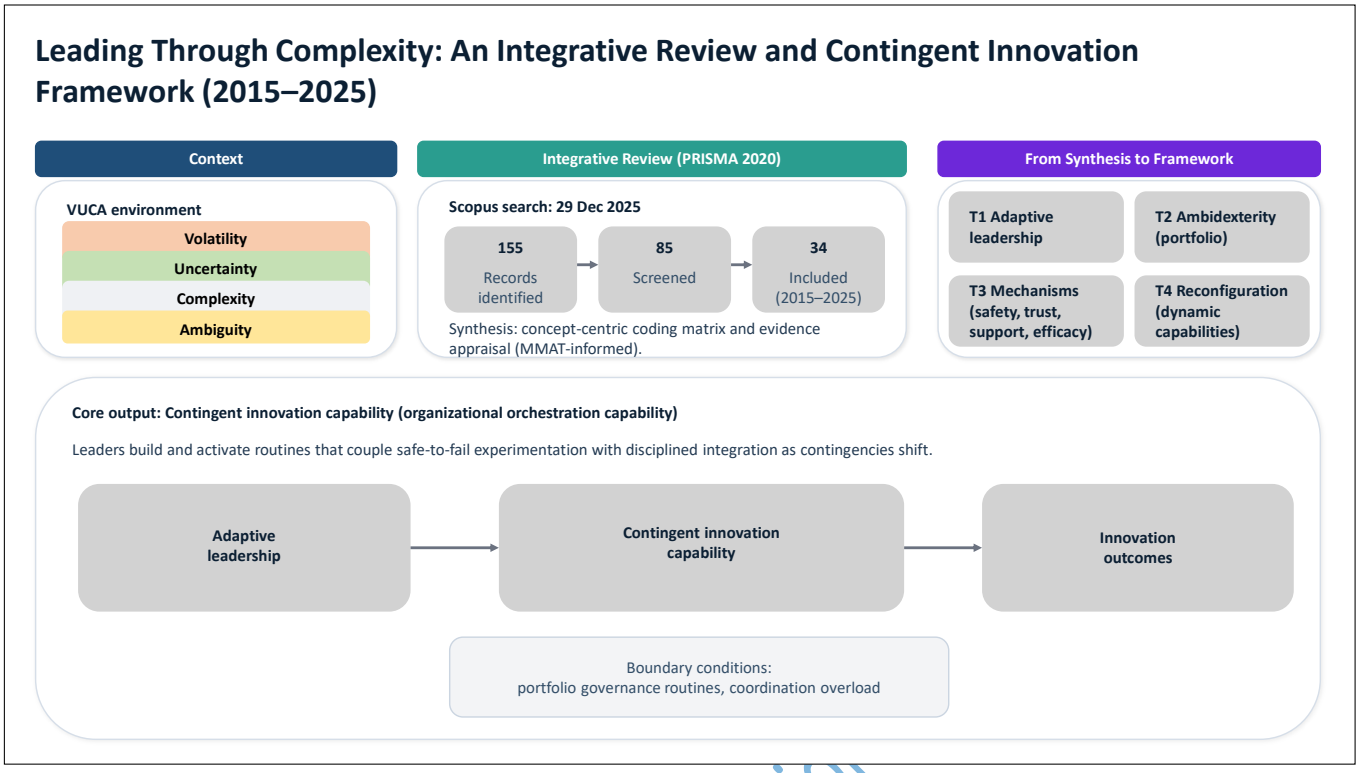
Practical implications: *The framework serves as a diagnostic and action tool and provides scenario-based leadership routines for balancing experimentation with disciplined alignment.*

Keywords: adaptive leadership; contingent innovation; complexity leadership; innovation portfolio management; organizational ambidexterity; dynamic capabilities

Plain Language Summary

In fast-changing environments, there is no single "best" innovation approach. Leaders need to read the situation, create safe conditions for learning and decide when to experiment and when to standardize. Reviewing recent research, this paper offers a four-proposition framework showing how adaptive leadership helps organizations shift their mix of innovation initiatives and resources to fit the context, while balancing psychological safety with disciplined alignment.

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1 Introduction

Organizations increasingly operate in environments characterized by volatility, uncertainty, complexity and ambiguity (VUCA) (Allio, 2025; Bennett and Lemoine, 2014). Technological disruption, geopolitical fragmentation, platform ecosystems and climate-related shocks amplify interdependence across supply chains and stakeholder networks (Chatzinikolaou and Vlados, 2022; Schoemaker and Day, 2021). In such contexts, innovation is simultaneously more necessary and more difficult: organizations must explore emerging opportunities while sustaining reliable execution under resource constraints and rapid digital change (Florek-Paszkowska and Ujwary-Gil, 2025; Majumdarr *et al.*, 2025).

Leadership research explains how leaders mobilize learning under uncertainty (sensemaking, psychological safety, trust), while innovation management explains how organizations configure innovation portfolios and reconfigure capabilities to adapt (Barbero *et al.*, 2017; Edmondson, 1999; Nagji and Tuff, 2012; Nguyen-Van *et al.*, 2025). Complementary work highlights how portfolio routines shape implementation of innovation strategy (Martinsuo *et al.*, 2024) and how control systems can influence portfolio agility and performance (Müller *et al.*, 2025; Roeth *et al.*, 2025).

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Yet these literatures are often treated separately: leadership studies emphasize micro-behaviors and enabling climates, while portfolio scholarship emphasizes decision rights, review cadences and resource (re)allocation processes. What remains under-specified is how leader behaviors translate into portfolio-level rebalancing and capability reconfiguration, and when those mechanisms help versus backfire under interdependence and overload. This review addresses that gap by synthesizing contemporary leadership and innovation-under-complexity research, while using established portfolio management scholarship to anchor the portfolio-governance component. Specifically, we define contingent innovation capability as an organizational orchestration capability (i.e., a set of routines for diagnosing shifting contingencies and adjusting the mix of innovation initiatives over time) and propose a proposition framework that links micro-behaviors to enabling conditions and portfolio reconfiguration, with portfolio governance routines and overload as boundary conditions. The guiding research questions (RQs) for this proposition framework are:

- RQ1: What adaptive leadership behaviors and roles are most consistently associated with innovation under complexity?
- RQ2: Through which psychological (relational) and capability (structural) mechanisms do these behaviors translate into innovative work behavior and organization-level innovation outcomes (e.g., ambidexterity and portfolio rebalancing where examined)?
- RQ3: Under what contingencies and boundary conditions do these mechanisms strengthen, weaken or backfire—and what does that imply for how contingent innovation differs from adjacent constructs?

Together, these RQs allow us to specify a tripartite mechanism—micro-behaviors/enabling conditions/portfolio reconfiguration—while making contingencies and failure modes explicit. To answer them, we adopt an integrative literature review because the focal phenomenon—leadership enabling innovation under complexity—cuts across leadership (e.g., Uhl-Bien and Arena, 2018), strategy (e.g., Hannevig, 2025), innovation management (e.g., Laufer *et al.*, 2025), digital transformation (e.g., Majumdarr *et al.*, 2025) and resilience (e.g., El-Gazar *et al.*,

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2025) research streams. Integrative reviews allow inclusion of diverse empirical designs and conceptual work and are appropriate for developing propositions and frameworks that integrate fragmented literatures (Snyder, 2019; Torraco, 2005; Whittemore and Knafl, 2005). We focus on 2015–2025 to capture contemporary evidence in the context of accelerating digital transformation and repeated systemic shocks, while using foundational pre-2015 works only to anchor definitions and boundary conditions.

We use "adaptive leadership" as an umbrella label for leadership that mobilizes learning and adjustment under complexity (Heifetz *et al.*, 2009; Uhl-Bien and Arena, 2018). "Complexity leadership" emphasizes enabling conditions for emergence and "adaptive space" in complex systems (Uhl-Bien *et al.*, 2007). "Leadership agility" focuses on rapid sensing and switching across competing demands (Gouda and Tiwari, 2024). "Ambidextrous leadership" refers to leader behaviors that deliberately balance "opening" (exploration) and "closing" (exploitation) behaviors (O'Reilly and Tushman, 2013).

The remainder of the paper is structured as follows. Section 2 details the integrative review methodology, including the selection process, quality appraisal and coding approach. Section 3 reports descriptive results and synthesizes four themes. Section 4 defines contingent innovation, clarifies conceptual boundaries, develops a four-proposition framework and provides an illustrative operationalization (Table 4 and four vignettes). It then discusses implications for practice and research and identifies limitations and prospects for future work. Section 5 concludes.

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2 Methodology

The search was conducted in Scopus on 29 December 2025. We searched titles, abstracts and keywords using a string including terms such as "adaptive leadership," "leadership agility," "complexity leadership," "ambidextrous leadership," "innovation," "innovative work behavior," "ambidexterity," VUCA and others (Table 1). We used Scopus because it provides broad multidisciplinary coverage and consistent export features that support transparent and replicable screening (Falagas *et al.*, 2008). To improve transparency and reproducibility, we

report the full query string and screening decisions and treat the review as a bounded Scopus-based synthesis rather than exhaustive coverage.

We limited the analytic sample to 2015–2025 to focus on contemporary evidence where complexity is shaped by digital transformation, ecosystem interdependence and repeated systemic shocks. Because adaptive leadership, complexity leadership and dynamic capabilities have earlier foundations, we cite seminal pre-2015 works only for conceptual anchoring (e.g., March, 1991; Teece, 2007; Uhl-Bien *et al.*, 2007) and do not treat them as part of the evidence base for the thematic synthesis.

Because our search string centers on leadership constructs, the portfolio management processes (e.g., portfolio formation, project selection, resource allocation and portfolio control) are less frequently examined directly in the leadership–innovation corpus. To ensure that "portfolio governance" is not an unsupported generalization, we treat it as a boundary condition and use the dedicated innovation portfolio management/project portfolio management literature to anchor the definition and operational indicators of portfolio governance routines (Aghajani *et al.*, 2023; Martinsuo *et al.*, 2024; Meifort, 2016; Müller *et al.*, 2025; Roeth *et al.*, 2025; Röth *et al.*, 2019; Si *et al.*, 2022; Stettina and Hörz, 2015). These sources inform concept specification and Table 2, but they are not counted among the 34 coded studies in the thematic synthesis.

We followed Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 reporting guidance (Page *et al.*, 2021) for search and selection transparency. The search returned 155 records; after filters, 85 were screened and 34 articles were included in the final synthesis (Table 1, Figure 1).

[Table 1 about here]

Table 1. Integrative review protocol and selection outcomes

Item	Specification
Review approach	Integrative literature review (concept-centric synthesis)
Search dates	29 December 2025
Databases / indexes	Scopus
Search string	TITLE-ABS-KEY (("adaptive leadership" OR "leadership agility" OR "complexity leadership" OR "ambidextrous leadership") AND (innovation OR "innovative work behavior" OR ambidexterity OR

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	exploration OR exploitation OR “dynamic capabilities” OR “digital transformation”) AND (complexity OR VUCA OR uncertainty OR turbulence OR crisis OR resilience)) AND PUBYEAR > 2014 AND PUBYEAR < 2027 AND (LIMIT-TO (DOCTYPE , “ar”)) AND (LIMIT-TO (LANGUAGE , “English”))
Inclusion criteria (summary)	Peer-reviewed journal articles in English (2015 onwards) examining adaptive/agile/complexity/ambidextrous leadership and its relationship to innovation outcomes/capabilities in complex, uncertain, turbulent or crisis contexts.
Exclusion criteria (summary)	Non-English records; publications before 2015; non-article document types; studies where leadership and/or innovation are peripheral; studies without an explicit complexity/uncertainty/turbulence/crisis/resilience context.
Sources included in synthesis (n)	34
Recency profile	29 sources (85%) published 2020–2025; 5 sources (15%) published 2015–2019.
Duplicates removed	0 (single Scopus export)
Other methods (citation chasing)	0 additional records
Reports not retrieved (full text unavailable)	2
Full-text exclusions (n)	4 (reasons reported in Figure 1)
Quality appraisal	MMAT-informed appraisal (empirical) and conceptual-rigor checklist (conceptual); used for evidence weighting

[Figure 1 about here]

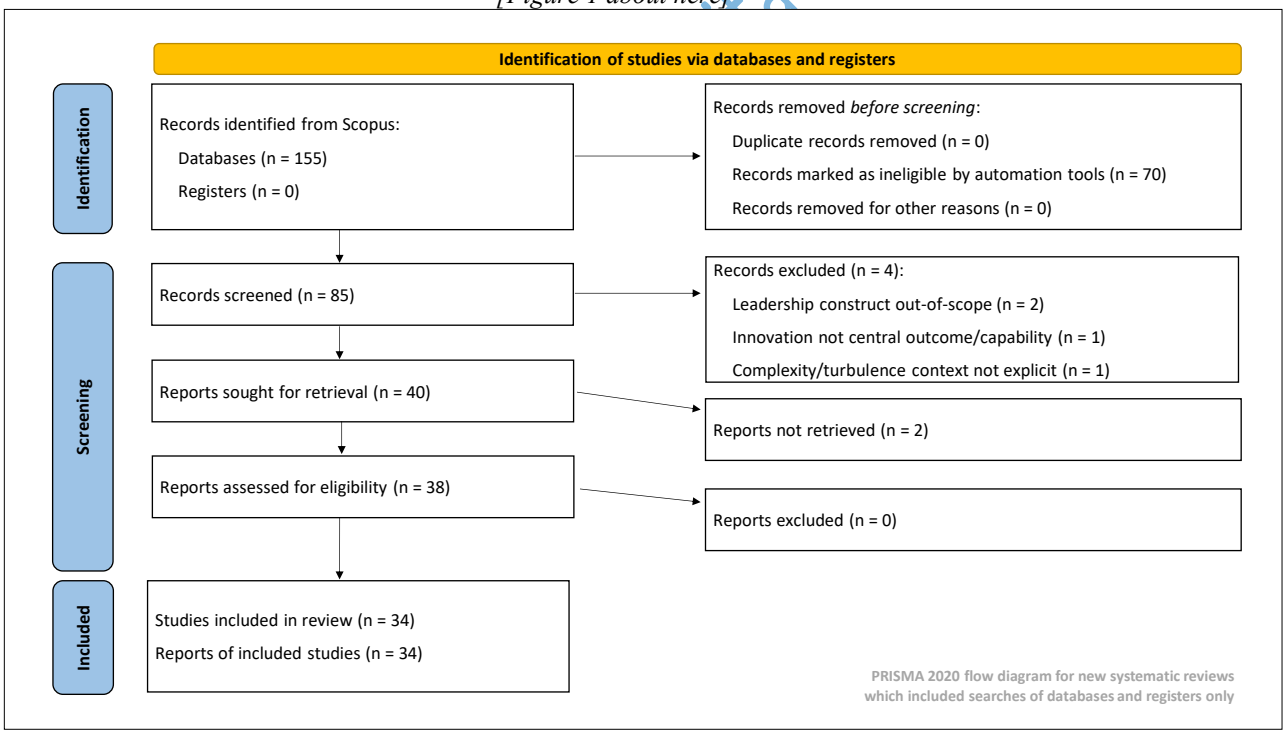


Figure 1. PRISMA 2020 flow diagram for the Scopus search (29 December 2025)

Given the heterogeneous mix of qualitative, quantitative, mixed-methods and conceptual papers, we added a “light-touch” quality appraisal step. For empirical studies, we used a Mixed Methods Appraisal Tool (MMAT)-informed checklist (Hong, 2018) to assess methodological

transparency and coherence. For conceptual papers, we assessed conceptual clarity, theoretical grounding and explicit boundary conditions. We did not exclude papers on quality grounds; instead, quality signals were used to weight evidence when deriving propositions (Gough, 2007).

Quality appraisal results (Supplementary File S1; download here) suggest the evidence base is generally methodologically transparent: 24 of 34 studies met at least 80% of applicable checklist items (heuristic high tier), 7 met 60–79% (moderate) and 3 met <60% (low). The most recurrent limitations flagged by the appraisal relate to quantitative surveys (e.g., limited ability to assess sample representativeness/selection bias and incomplete reporting of missing data/non-response) and, for some conceptual papers, variability in how explicitly the evidence base and boundary conditions are documented. These signals were used to calibrate the strength of thematic statements and to motivate the paper's emphasis on contingencies and failure modes rather than universal effects (Lewin *et al.*, 2018).

Synthesis proceeded in three steps. First, we built a concept-centric coding matrix and applied deductive codes aligned with RQ1–RQ3 (leadership behaviors/roles; mechanisms; contingencies/boundary conditions; outcomes and levels of analysis). All full texts were read in full and coded into the matrix. Second, we conducted inductive coding to capture emergent tensions (e.g., exploration–execution conflicts) and non-linear dynamics (e.g., overload). Third, we compared codes across studies and levels of analysis to develop integrative themes and propositions. Disagreements in coding and theme boundaries were resolved through iterative refinement of the codebook, with an audit trail maintained to document decisions. Appendix A presents the audit trail; Appendix B explains the matrix and provides an illustrative coded example; Appendix C summarizes the included studies. The complete per-study quality appraisal scoring (MMAT-informed and conceptual-rigor) and supporting evidence pointers are provided as Supplementary File S1. The complete 34-study full-text coding matrix is provided as Supplementary File S2 (download here) to support auditability of the synthesis.

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3 Results: Evidence Base and Thematic Synthesis

The included studies span management and organization studies, public administration, higher education, healthcare and information systems, reflecting the cross-sector character of complexity-driven innovation challenges (Callens, 2024; Cousins *et al.*, 2023; Ho *et al.*, 2025; Laufer *et al.*, 2025; Majumdarr *et al.*, 2025). Across studies, innovation is operationalized at multiple levels: employee innovative work behavior, team innovation, organizational innovation performance and innovation-related capabilities such as ambidexterity and dynamic capabilities. However, few of the included leadership studies examine innovation portfolio management processes (e.g., portfolio design, selection, resource allocation and control) directly; portfolio logic appears primarily through the use of ambidexterity as an organization-level outcome. A consistent pattern is that leadership constructs are increasingly treated as behavioral configurations that enable switching between exploration and exploitation rather than as static traits. Table 2 summarizes the main themes and representative evidence.

[Table 2 about here]

Table 2. Thematic synthesis: core themes, mechanisms and indicative evidence

Theme	Summary (what the evidence suggests)	Sources (coded corpus)
1. Adaptive leadership behaviors in complex systems	Adaptive leadership is enacted through sensemaking/framing, enabling experimentation and orchestrating alignment under uncertainty. Several sources argue that adaptiveness is most effective when paired with explicit coordination boundaries and decision rights to reduce fragmentation; however, direct empirical evidence on this boundary-setting mechanism is uneven in the reviewed corpus.	(Biggadike *et al.*, 2022; Callens, 2024; Carroll *et al.*, 2024; Fernández Fernández, 2022; Ho *et al.*, 2025; Jiang *et al.*, 2023; Mendes *et al.*, 2016; Schophuizen *et al.*, 2023; Schulze and Pinkow, 2020; Tsai *et al.*, 2019; Uhl-Bien, 2021)
2. Ambidexterity as a portfolio-level* outcome (and portfolio governance)	Balancing/sequencing exploration and exploitation is the recurrent strategic tension under turbulence. In the leadership-focused corpus, ambidexterity is typically treated as an organization-level (portfolio) outcome—the evolving mix of exploratory and exploitative initiatives—rather than as a single-project decision. Direct evidence on innovation portfolio management processes (portfolio design, resource allocation, portfolio control) is comparatively limited; portfolio-governance routines are therefore anchored in dedicated innovation portfolio management scholarship.	(Diesel and Scheepers, 2019; Gouda and Tiwari, 2024; Hannevig, 2025; Nguyen-Van *et al.*, 2025; Schindler *et al.*, 2024; Zuraik *et al.*, 2020)
3. Mechanisms translating leadership into innovation	Psychological safety and vertical trust increase employees' willingness to experiment; organizational support/resources (e.g., infrastructure, communication quality and managerial support) and creative self-efficacy increase perceived capability to innovate. These mechanisms become especially important under crisis and high uncertainty.	(Bernards, 2025; El-Gazar *et al.*, 2025; Indrianti *et al.*, 2024; Khan *et al.*, 2022, 2025; Laufer *et al.*, 2025; Lestari *et al.*, 2024; Majumdarr *et al.*, 2025; Oeij *et al.*, 2016)

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4. Dynamic capabilities and reconfiguration	Dynamic capabilities (sensing, seizing, transforming) explain how organizations repeatedly reconfigure resources and routines so that experiments scale into innovative outcomes. Leadership behaviors shape attention, investment timing and learning loops; governance routines condition whether reconfiguration reduces or amplifies coordination costs.	(Ajith *et al.*, 2025; Anand *et al.*, 2024; Björck *et al.*, 2025; Cousins *et al.*, 2023; Florek-Paszkowska and Ujwary-Gil, 2025; Schoemaker and Day, 2021; Syamsir *et al.*, 2025; Uhl-Bien and Arena, 2018)

(*Note: For portfolio-level anchors, see Aghajani *et al.*, 2023; Martinsuo *et al.*, 2024; Meifort, 2016; Müller *et al.*, 2025; Roeth *et al.*, 2025; Röth *et al.*, 2019; Si *et al.*, 2022; Stettina and Hörz, 2015)

Integrating the "quality appraisal" with the thematic synthesis adds an additional confidence lens. Overall, evidence supporting the four themes is dominated by studies appraised as "high" quality (24/34), with Themes 1–2 drawing on high-to-moderate evidence (no low-tier papers), whereas the few low-tier papers appear in Themes 3–4 and are primarily quantitative surveys where representativeness and missing-data/non-response issues were not fully assessable (Supplementary File S1). We therefore interpret the psychological-relational mechanisms and the reconfiguration pathway as robustly signposted but especially sensitive to measurement and context, reinforcing the contingent (non-universal) framing of leadership effects under complexity.

Theme 1 primarily addresses RQ1 (adaptive leadership behaviors and roles). Themes 2 and 4 address RQ2 by synthesizing the portfolio/ambidexterity logic and the reconfiguration pathway via dynamic capabilities, and they inform RQ3 by clarifying when portfolio shifting is feasible or constrained. Theme 3 addresses RQ2 by synthesizing the psychological and relational mechanisms that translate leadership into innovative work behavior. The cross-theme synthesis surfaces RQ3 tensions and failure modes (e.g., safety versus discipline; overload) that motivate contingent propositions.

Theme 1 – Adaptive leadership as sensemaking and enabling in complexity: Across the corpus, adaptive leadership is conceptualized as the capacity to interpret weak signals, reframe problems and mobilize learning (Jiang *et al.*, 2023; Schulze and Pinkow, 2020). Adaptive leadership emphasizes getting people to confront reality and experiment with new practices (Heifetz *et al.*, 2009). Conceptual discussions of ambiguous innovation further underline the role of leadership framing and mindful resilience under uncertainty (Fernández Fernández, 2022). Earlier CLT-oriented (complexity leadership theory) accounts also clarify how adaptive,

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enabling and administrative leadership roles combine to promote learning and innovation (Mendes *et al.*, 2016), including in socio-technical adoption domains such as learning analytics (Tsai *et al.*, 2019).

A critical subtlety, however, is that adaptiveness is not universally beneficial. Under high interdependence, indiscriminate experimentation can create fragmentation, role ambiguity and change fatigue (Biggadike *et al.*, 2022; Carroll *et al.*, 2024; Jiang *et al.*, 2023). The leadership challenge is therefore to couple experimentation with boundaries, priorities and integration mechanisms. This subtlety becomes central once leadership is integrated with innovation strategy, because innovation requires both opening behaviors (idea generation, experimentation) and closing behaviors (implementation discipline, coordination).

Theme 2 – Ambidexterity frames innovation under complexity as an exploration–exploitation portfolio problem: In the leadership–innovation corpus, the dominant strategic framing is ambidexterity—the need to balance or sequence exploration and exploitation under turbulence (Diesel and Scheepers, 2019; Hannevig, 2025; O'Reilly and Tushman, 2013). When studied at the organizational level, this balance is typically discussed as a portfolio-level outcome (i.e., the evolving mix of exploratory and exploitative initiatives) rather than as a single-project decision. However, direct empirical attention to innovation portfolio management processes (portfolio design, project selection, resource allocation and control) is limited in the leadership-focused evidence base. Dedicated innovation portfolio and project portfolio management research provides complementary evidence on the governance routines that enable portfolio agility (Martinsuo *et al.*, 2024; Meifort, 2016; Müller *et al.*, 2025; Roeth *et al.*, 2025; Röth *et al.*, 2019; Si *et al.*, 2022; Stettina and Hörz, 2015).

Within our sample, ambidexterity is linked to climate and behavioral mechanisms (Diesel and Scheepers, 2019), strategic leadership transitions and governance choices (Hannevig, 2025) and disruptive innovation settings that require deliberate handling of exploration–exploitation tensions (Schindler *et al.*, 2024; Zuraik *et al.*, 2020). Recent sustainability-oriented work further extends ambidexterity into green innovation contexts under VUCA conditions (Nguyen-Van *et*

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al., 2025), while human capital research links ambidextrous leadership to agility and engagement outcomes that support innovation execution (Gouda and Tiwari, 2024).

Theme 3 – Mechanisms linking leadership to innovative behavior: Beyond direct effects, the corpus shows leadership influences innovation via psychological and relational pathways that shape whether employees will take interpersonal and performance risks, particularly under uncertainty (Bernards, 2025; Laufer *et al.*, 2025; Oeij *et al.*, 2016). Psychological safety and vertical trust recur as central mechanisms (Edmondson, 1999). For example, leaders' trust-building practices can enable experimentation and adoption of educational technology innovations in universities (Laufer *et al.*, 2025), while cognitive uncertainty can suppress daily innovative work behavior unless moderated by ambidextrous leadership (Bernards, 2025). Complementary evidence links ambidextrous leadership to innovative performance through self-efficacy and resilience resources (El-Gazar *et al.*, 2025; Khan *et al.*, 2022), including in entrepreneurial settings where mindfulness supports innovation capability under turbulence (Indrianti *et al.*, 2024).

Additional evidence reinforces the role of knowledge and relational mechanisms: knowledge sharing mediates the relationship between clan culture, trust/resilience resources and innovative work behavior in village-owned enterprises (Lestari *et al.*, 2024). Research on innovation teams similarly links mindful infrastructure with innovation resilience behavior via team psychological safety, learning and voice, alongside complexity leadership (Oeij *et al.*, 2016). In academic settings during coronavirus disease 2019 (COVID-19), adaptive leadership is associated with innovative behavior and readiness for change, underscoring crisis as an amplifier of mechanism salience (Khan *et al.*, 2025).

This mechanism evidence strengthens two implications for theory building. First, leadership effects on innovation in complex contexts are unlikely to be purely direct; they are transmitted through micro-level psychological states and meso-level relational climates. Second, mechanism strength is itself contingent: for example, trust becomes more critical in high-risk environments, while organizational support and efficacy become more salient in knowledge-intensive contexts where autonomy and expertise shape innovation behavior.

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Theme 4 – Dynamic capabilities and changing how we do things: Building on the dynamic capabilities view (Teece, 2007), the evidence base emphasizes leaders' role in sensing emerging threats/opportunities, seizing them through timely investment and partnership choices and continuously reconfiguring resources and routines. In the themes, ambidexterity (Theme 2) frames the exploration–exploitation tension (often treated as a portfolio-level outcome), whereas dynamic capabilities specify the reconfiguration processes that make repeated shifts in the mix of innovation initiatives feasible over time. Across crisis and disruption settings, case-based evidence highlights rapid learning loops and resource redeployment as pivotal for translating local experimentation into organization-level adaptation (Ajith *et al.*, 2025; Björck *et al.*, 2025; Cousins *et al.*, 2023).

Conceptual syntheses further foreground turbulence, digital transformation and VUCA conditions as contexts where reconfiguration capacity becomes a strategic bottleneck; leadership agility and enabling conditions shape whether sensing–seizing–transforming cycles reduce or amplify coordination costs (Schoemaker and Day, 2021; Syamsir *et al.*, 2025; Uhl-Bien and Arena, 2018). This is especially salient when innovation and sustainability expectations converge with digital transformation and broader ecosystem turbulence (Anand *et al.*, 2024; Florek-Paszkowska and Ujwary-Gil, 2025).

Several studies imply that the mechanisms supporting exploration can create friction with disciplined execution (Biggadike *et al.*, 2022; Jiang *et al.*, 2023; Schindler *et al.*, 2024; Uhl-Bien and Arena, 2018). For example, psychological safety that encourages voice and experimentation (Theme 3) may conflict with the need for rapid alignment and standardization during scale-up (Theme 1). Similarly, portfolio-level ambidexterity (Theme 2) requires integrating exploratory and exploitative work, but dynamic capability reconfiguration (Theme 4) can amplify coordination demands if governance routines and decision rights are weak. These tensions are central to contingent innovation because they explain why "more" of a leadership behavior is not always better and why boundary conditions matter.

To synthesize and define contingent innovation, the themes converge on a clarifying point: "contingency" in innovation is not merely the observation that context matters. Innovation

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under complexity is contingent because the appropriate mix of innovation initiatives (i.e., the innovation portfolio), governance routines and behavioral stance changes across time and situations (Martinsuo *et al.*, 2024; Meifort, 2016; Si *et al.*, 2022). We therefore define contingent innovation capability as the organization's ability to diagnose shifting contingencies and adjust that mix over time by coupling experimentation with enabling conditions and integration routines; adaptive leadership behaviors help develop and activate this capability.

Bringing RQ1–RQ3 together, the synthesis suggests that adaptive leadership enables contingent innovation through two interdependent pathways: psychological-relational mechanisms (psychological safety/trust and organizational support/self-efficacy) that facilitate risk-taking and experimentation, and capability-structural mechanisms (dynamic capabilities and ambidexterity) that translate local learning into organization-wide innovation reconfiguration. These mechanisms are amplified under conditions of high environmental complexity, with organizational design and governance routines serving as critical boundary conditions that determine whether adaptive leadership produces productive innovation or costly fragmentation.

4 Discussion

4.1 Differentiating Contingent Innovation from Adjacent Constructs

Contingent innovation overlaps with contingency theory, ambidexterity, dynamic capabilities and innovation portfolio management, but it is not reducible to any one of them (Table 3). Contingency theory emphasizes fit between environment and organizational arrangements (Lawrence and Lorsch, 1967), but often remains macro-level and descriptive regarding the leadership processes that generate and revise fit over time. Ambidexterity specifies the core innovation tension (exploration vs exploitation) and alternative designs for balancing it (O'Reilly and Tushman, 2013), but can under-specify the leader behaviors and enabling conditions through which balancing is enacted in practice. Dynamic capabilities focus on sensing, seizing and transforming (Teece, 2007), but require clearer micro-foundations that explain how managerial action builds and renews reconfiguration capacity (Barbero *et al.*,

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2017). Innovation portfolio management focuses on portfolio design, selection, resource allocation and control routines that shape the innovation initiative mix (Meifort, 2016; Röth *et al.*, 2019; Si *et al.*, 2022; Stettina and Hörz, 2015). We position contingent innovation capability as an organizational orchestration capability that leaders help develop and activate, connecting these elements by treating the innovation portfolio as the strategic object (Meifort, 2016; Nagji and Tuff, 2012)—and by making governance routines and overload explicit boundary conditions that determine whether adaptiveness yields coordinated innovation rather than fragmentation (Martinsuo *et al.*, 2024; Roeth *et al.*, 2025).

[Table 3 about here]

Table 3. Conceptual boundaries: contingent innovation versus adjacent constructs

Construct	Core focus	Typical limitation in prior work	How contingent innovation differs
Contingency theory	Environment–structure fit	Often descriptive; limited specification of leadership micro-processes	Specifies leadership behaviors/mechanisms through which fit is enacted and revised over time
Ambidexterity	Balancing/sequencing exploration and exploitation	May over-emphasize structural solutions; leadership mechanisms often coarse	Treats ambidexterity as a portfolio outcome; specifies leadership "switching" behaviors and psychological mechanisms that enable balance
Dynamic capabilities	Sensing–seizing–transforming (reconfiguration capacity)	Micro-foundations sometimes underspecified; weak links to employee-level innovation mechanisms	Connects reconfiguration routines to leader behaviors and employee-level enabling conditions (trust/safety, support/resources, creative self-efficacy)
Innovation portfolio management	Portfolio formation/design, selection, resource allocation and control of innovation initiatives	Often emphasizes formal processes; leadership micro-behaviors and enabling climates may be implicit	Clarifies how leadership behaviors build/activate portfolio governance routines under shifting contingencies and how those routines can prevent fragmentation and overload
Adaptive/complexity leadership	Learning, emergence and adaptation in complex systems	May treat innovation as unitary outcome; limited linkage to portfolio governance	Makes portfolio choices, governance routines and boundary conditions explicit as part of leadership-driven orchestration

From this position, contingent innovation advances beyond descriptive "fit" accounts (Vlados and Chatzinikolaou, 2025), specifying a multilevel mechanism that links leader behaviors and enabling conditions to organization-level innovation outcomes (including portfolio-level balancing) and reconfiguration processes. Specifically, it treats exploration–exploitation balancing as a dynamic portfolio problem (March, 1991; Meifort, 2016; O'Reilly and Tushman,

2013) and explains how shifts in the innovation initiative mix scale into organization-level change through repeated sensing–seizing–transforming cycles (Si *et al.*, 2022; Teece, 2007). The distinctive contribution is therefore not a new label for "context matters," but an integrative specification of how leadership-enabled routines (sensemaking, bounded experimentation and disciplined alignment) translate into portfolio ambidexterity and capability reconfiguration, contingent on governance and integration discipline (Martinsuo *et al.*, 2024; Roeth *et al.*, 2025).

4.2 Propositions and Operationalization

The following propositions (Figure 2) translate the synthesis into testable statements. They are written at a general level but can be operationalized at individual, team and organizational levels depending on research design. Because direct evidence on innovation portfolio management processes is limited in the leadership-focused corpus, Propositions 2 and 4c treat portfolio governance routines as a theoretically grounded boundary condition and use innovation portfolio management/project portfolio management research to anchor the definition of governance routines and observable indicators (Martinsuo *et al.*, 2024; Meifort, 2016; Müller *et al.*, 2025; Roeth *et al.*, 2025; Röth *et al.*, 2019; Si *et al.*, 2022; Stettina and Hörz, 2015).

[Figure 2 about here]

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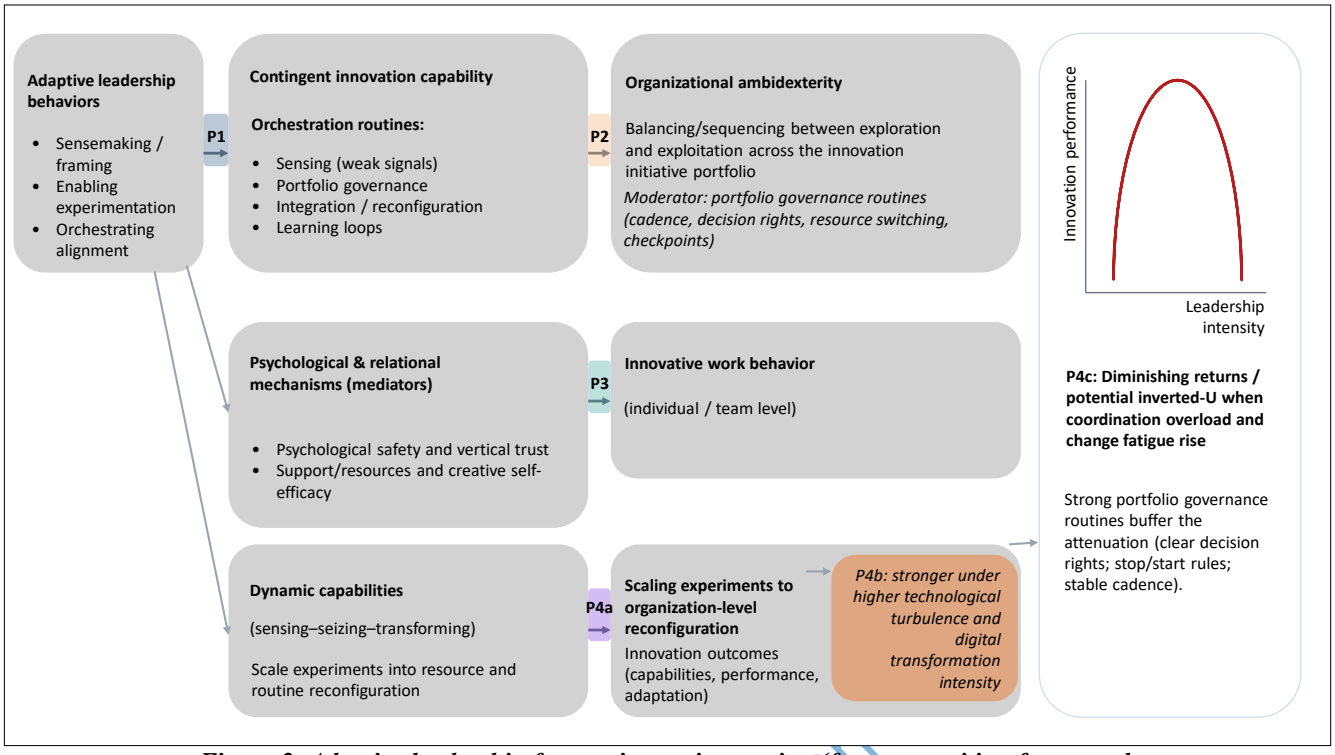


Figure 2. Adaptive leadership for contingent innovation (four-proposition framework; Proposition 4a–4c specify subcomponents of Theme 4)

Proposition 1 (Theme 1: adaptive leadership behaviors; multilevel): Adaptive leadership behaviors—sensemaking/framing, enabling experimentation and orchestrating alignment—are positively associated with the development and activation of the organizational routines that constitute contingent innovation capability (e.g., sensing routines, bounded experimentation processes, portfolio routines, integration mechanisms and learning loops) as contingencies shift.

Proposition 2 (Theme 2: organizational ambidexterity as a portfolio-level outcome; organizational level): Adaptive leadership behaviors are positively associated with organizational ambidexterity (dynamic balancing/sequencing of exploration and exploitation across an evolving innovation initiative portfolio). This association is stronger when portfolio governance routines (portfolio review cadence, resource-switching rules, integration checkpoints and clear decision rights) are more developed.

Proposition 3 (Theme 3: psychological mechanisms; individual/team level): The relationship between adaptive leadership behaviors and innovative work behavior is mediated by (a) psychological safety and vertical trust and (b) organizational support/resources (e.g.,

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infrastructure, communication quality, managerial support and protected time) and creative self-efficacy. The indirect effect via psychological safety is strengthened when experimentation is coupled with disciplined alignment (clear priorities and feedback loops); in the absence of alignment, psychological safety increases the risk of fragmented experimentation, while alignment without safety suppresses experimentation.

Proposition 4a (Theme 4: dynamic capabilities; multilevel): Dynamic capabilities (sensing, seizing and transforming) mediate the relationship between adaptive leadership behaviors and the scaling of experiments into organization-level reconfiguration.

Proposition 4b (Theme 4: boundary conditions; multilevel): The dynamic-capabilities mediation in Proposition 4a is stronger under higher technological turbulence and greater digital transformation intensity.

Proposition 4c (Theme 4: overload; proposed extension): Adaptive leadership intensity may exhibit diminishing returns (and, in some contexts, an inverted-U pattern): beyond an overload threshold, coordination costs and change fatigue attenuate innovation performance. Strong portfolio governance routines (clear decision rights, stop/start rules and a stable portfolio review cadence) should buffer this attenuation; modular architectures are a plausible additional buffer (theory-driven extension) that warrants explicit testing.

4.3 Implications

For practitioners, the framework is most actionable when applied to a specific innovation episode. The starting point is to diagnose the dominant contingency (e.g., crisis intensity and interdependence; digital turbulence; regulatory constraint; or low psychological safety) and choose an innovation stance—exploit, explore or a sequenced hybrid. Table 4 translates Propositions 1–4 into a scenario-based "if–then" guide by specifying (i) observable leadership routines that enable both learning and alignment and (ii) guardrails that signal fragmentation or overload.

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[Table 4 about here]

Table 4. Scenario-based leadership routines and guardrails aligned with Themes 1–4 and Propositions 1–4

Dominant contingency	Innovation stance	Leadership routines (observable)	Guardrails / failure signals	Framework linkage
High-crisis, high interdependence; reliability critical	Exploit and targeted safe-to-fail exploration	Daily/weekly sensemaking cadence; triage decisions; rapid, blameless after-action reviews and structured speak-up; micro-experiments in bounded zones; explicit escalation and decision rights; resource-switching rules and integration checkpoints; close-out routines for scale-up.	Rising workarounds; role ambiguity; conflicting priorities; fatigue and backlog growth	Primarily P1 & P2; also, P3 & P4c
High digital turbulence; platform/ecosystem change	Explore and modular scaling	Weak-signal scanning; experiment portfolio with clear hypotheses; modular architectures*; dynamic resource reallocation; cross-functional learning loops; stage-gate redesign for fast pivots; explicit integration checkpoints and decision rights for scale/pause/stop.	Experiment sprawl; integration delays; technical debt; local wins not scaling	Primarily P4a & P4b; also, P1 & P2 (and P4c for overload risks)
Public accountability / high regulation; mission continuity	Hybrid (innovation within constraints)	Define non-negotiables; co-design with stakeholders; transparent risk log; clear decision rights and compliance sign-off checkpoints; protected learning time; standardize once proven; communicate trade-offs and legitimacy rationale; safe speak-up channel for compliance/ethics risks.	Compliance breaches; stakeholder distrust; innovation perceived as discretionary or unsafe	Primarily P1 & P2; also, P3
Low psychological safety / low trust; high interpersonal risk; cross-functional interdependence	Explore within guardrails (voice-first)	Leader role-modelling of fallibility; structured speak-up rounds; blameless after-action reviews; small reversible experiments with clear hypotheses and stop rules; coaching and resources to build self-efficacy; clear priorities and fast feedback loops; recognition of learning behaviors.	Silence in meetings; error hiding; idea hoarding; retaliation fears; rising turnover; experiments proliferate without convergence or decisions.	Primarily P3; also, P1 & P2

Operationally, Table 4 can be used as a short-cycle diagnostic in three steps. First, establish a sensemaking cadence that surfaces weak signals, clarifies "non-negotiables," and frames the innovation problem in terms of the dominant contingency. Second, bound experimentation by specifying decision rights, integration checkpoints and resource-switching rules (portfolio governance), so that exploration remains coherent and scalable. In parallel, pair structured speak-up and blameless after-action reviews with clear priorities and fast feedback loops so that psychological safety translates into disciplined learning rather than fragmented experimentation. Third, monitor early warning indicators—role ambiguity, experiment sprawl,

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integration delays, rising workarounds and change fatigue—and tighten alignment routines before the system tips into coordination overload.

For researchers, the routines in Table 4 can be treated as observable indicators of adaptive leadership enactment and as proxies for contingent innovation capability, while the propositions specify the mediators and boundary conditions that should be measured explicitly. Psychological safety, vertical trust, organizational support/resources (e.g., infrastructure, communication quality, managerial support, protected time) and creative self-efficacy can be operationalized at individual/team levels; innovative work behavior can capture near-term experimentation outcomes; portfolio ambidexterity and resource reallocation can capture shifts in the innovation initiative mix (portfolio shifts); and dynamic capabilities and reconfiguration outcomes can be tracked longitudinally (e.g., changes in routines, structures and resource deployments). Proposition 4c further directs attention to overload measurement (coordination costs, backlog growth, change fatigue) and to testing non-linear effects.

Below we provide illustrative vignettes that operationalize the scenario-based routines and guardrails in Table 4. The vignettes are stylized examples intended to demonstrate how Propositions 1–4 translate into observable leadership routines, mediating mechanisms and boundary-condition "failure signals" across different contingency profiles. They are included for interpretive clarity and teaching purposes and should not be read as part of the empirical evidence base of the review.

Illustrative vignette 1 (high-crisis, high interdependence; reliability critical): A hospital responds to a cyber incident while piloting an artificial intelligence (AI)-enabled triage workflow. Leaders institute a brief daily sensemaking cadence to triage what must remain reliable versus where bounded experimentation is permissible (Proposition 1). They authorize micro-experiments only in clearly defined zones, with explicit escalation paths, decision rights, integration checkpoints and resource-switching rules, and they use close-out routines (scale, pause, stop) to prevent experiment sprawl and ensure that learning integrates and scales (Proposition 2). Rapid, blameless after-action reviews and structured speak-up routines reinforce psychological safety and vertical trust, while targeted support (time, rapid training,

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access to technical expertise) strengthens self-efficacy for adapting workflows (Proposition 3). Leaders watch for overload signals—conflicting priorities, role ambiguity, fatigue, rising workarounds and backlog growth—and tighten portfolio governance, integration checkpoints and alignment routines before coordination costs suppress learning and performance (Proposition 4c).

Illustrative vignette 2 (high digital turbulence; platform/ecosystem change): A manufacturing medium-sized enterprise (SME) faces volatile demand and supplier disruption while migrating to a new enterprise resource planning (ERP) system and expanding Internet of Things (IoT) instrumentation. Leaders build weak-signal scanning into weekly reviews, maintain a visible experiment portfolio with hypotheses and stop rules and redesign stage-gates to support fast pivots without losing integration discipline (Proposition 1). Dynamic resource reallocation and cross-functional learning loops operationalize sensing–seizing–transforming cycles that help scale local experiments into organization-level reconfiguration (Proposition 4a), and the value of these cycles is amplified under high technological turbulence and digital transformation intensity (Proposition 4b). As a plausible design extension, modular architectures (or other decoupling mechanisms) can reduce integration costs and technical debt accumulation during scale-up, complementing governance routines (integration checkpoints, decision rights, scale/pause/stop rules) that determine whether local experiments translate into portfolio rebalancing rather than disconnected pilots (Proposition 2). Guardrails include experiment sprawl, integration delays, technical debt accumulation and local wins that do not scale—signals that governance and alignment routines need tightening and that the system may be approaching coordination overload (Proposition 4c).

Illustrative vignette 3 (public accountability/high regulation; mission continuity): A municipal agency digitizes permitting under strict compliance, transparency and equity requirements. Leaders define non-negotiables up front; co-design bounded pilots with key stakeholders and maintain a transparent risk log that makes trade-offs explicit (Proposition 1). They protect learning time during exploration and then standardize workflows once reliability and legitimacy are demonstrated, coupling psychological safety (voice and error reporting) with

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disciplined alignment (clear priorities and feedback loops) (Proposition 3). Portfolio governance routines specify which services can be piloted, what evidence is required to scale, who has decision rights and where compliance sign-off checkpoints apply, preventing ad hoc experimentation that would undermine public trust (Proposition 2). Guardrails include compliance breaches, stakeholder distrust and narratives that innovation is discretionary or unsafe—signals that governance, communication and integration discipline require recalibration before coordination costs and legitimacy risks accumulate (Proposition 4c).

Illustrative vignette 4 (low psychological safety/low trust; experimentation stalls): A cross-functional product team is asked to redesign a customer onboarding journey, but prior failures have produced a blame climate and employees avoid raising risks. Leaders start by restoring psychological safety and vertical trust: they introduce structured speak-up rounds, run blameless retrospectives and publicly reward "early warning" behaviors (Proposition 3). They also reframe the work into a small set of testable hypotheses with clear priorities and feedback loops, clarifying what must remain stable versus what can be varied (Proposition 1). In parallel, they provide tangible support—time buffers, coaching, lightweight prototyping tools and access to expertise—to strengthen creative self-efficacy and reduce perceived personal risk of experimentation (Proposition 3). As safety rises, the team increases the rate and quality of small, reversible experiments with explicit stop rules while maintaining alignment through explicit decision rights, integration checkpoints and shared priorities (Proposition 2). Guardrails include silence in meetings, superficial agreement, risk concealment, declining idea flow and experiments proliferating without convergence—signals that safety is not yet translating into disciplined learning, and that additional support and boundary setting are required (Propositions 1–3).

4.4 Limitations and Prospects

Despite a transparent protocol, the review has limitations that should bound interpretation. About method limitations, first, the synthesis is designed for theory building and cannot support causal claims or effect-size estimates. Second, the evidence base is bounded to Scopus-indexed,

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English-language journal articles (2015–2025); relevant work in other databases, books and practitioner outlets may be under-represented. Third, the included studies remain uneven across sectors and levels of analysis and, in some survey-based papers, representativeness and missing-data/non-response issues limit generalizability. Fourth, the leadership-centered query yielded fewer studies that examine innovation portfolio management processes directly; accordingly, portfolio governance routines in the framework are grounded in adjacent innovation portfolio management/project portfolio management scholarship rather than treated as strongly evidenced findings from the coded corpus.

Concerning framework limitations, first, the core construct—contingent innovation capability—requires further measurement development to distinguish it empirically from adjacent constructs (e.g., ambidexterity and dynamic capabilities). Second, the framework specifies key mechanisms and boundary conditions but abstracts from detailed temporal sequencing (when and how quickly leaders should switch between exploration and exploitation). Third, Proposition 4c proposes an overload threshold (proposed extension), but the location of that threshold is likely context-dependent and requires empirical calibration.

These limitations point to seven priorities for future research. (1) Replicate and extend the review using multi-database searches and broader source types to test the robustness of the thematic patterns. (2) Develop and validate measures of contingent innovation capability, portfolio governance routines and overload signals. (3) Use multilevel research designs that connect leader routines to innovative work behavior through psychological safety/trust and support/self-efficacy. (4) Treat the innovation portfolio as a focal unit of analysis by operationalizing portfolio composition (e.g., horizons, risk/uncertainty profiles, exploration–exploitation balance), decision rules and resource allocation logics, and by examining how leadership shapes portfolio-level choices and trade-offs. (5) Employ longitudinal and event-based approaches to capture portfolio rebalancing and reconfiguration over time, especially during crises and digital transformation programs. (6) Test boundary conditions explicitly (e.g., crisis intensity, technological turbulence, regulation, interdependence) and model non-linear

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effects (Propositions 4b and 4c). (7) Examine how ecosystem partners and public legitimacy demands shape governance routines and the scaling of experiments into reconfiguration.

Consistent with the quality appraisal (Supplementary File S1), several quantitative studies relied on bounded or convenience samples and did not fully address selection bias/representativeness, missing data or non-response/attrition. These limitations constrain generalizability and may inflate reported associations in parts of the quantitative evidence base. In addition, some conceptual papers varied in how explicitly they documented their evidence base and analytic steps, which reinforces interpreting the framework as an integrative proposition to be tested (rather than as a cumulative causal estimate).

5 Conclusion

This integrative review answers RQ1 by showing that adaptive leadership is most consistently operationalized as sensemaking/framing, enabling experimentation and orchestrating alignment as ideas move toward implementation. It answers RQ2 by indicating that leadership influences innovation primarily through (a) psychological-relational enabling conditions (trust, psychological safety, organizational support/resources and creative self-efficacy) and (b) organization-level reconfiguration processes (ambidexterity and dynamic capabilities) that help local learning scale into shifts in the innovation initiative mix. It answers RQ3 by highlighting that these mechanisms are contingent on environmental turbulence and governance routines and can backfire via fragmentation, coordination overload and change fatigue. The resulting construct of contingent innovation captures this multilevel and contingent (and sometimes paradoxical) logic.

By clarifying conceptual boundaries and advancing a focused set of four testable propositions, the paper offers a research agenda for studying leadership-driven innovation reconfiguration across contingencies such as crisis intensity, technological turbulence and regulatory constraint. For practitioners, the framework translates into observable routines for diagnosing context, designing a portfolio stance and balancing experimentation with disciplined alignment (Table 4). Future empirical work should test the propositions with

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multilevel, longitudinal designs and identify where adaptive leadership becomes counterproductive.

Future work can now move from broad calls for "agile" leadership to empirical tests of the framework's contingent mechanisms by tracking how sensemaking, experimentation and alignment interact with psychological safety, governance routines and dynamic capabilities across disruption episodes. For practitioners, the actionable message is to treat innovation as an evolving portfolio: create protected space for experimentation, but pair it with clear decision rights, integration routines and reconfiguration capabilities so that learning scales without tipping the system into overload.

Appendix A. Audit Trail of the Synthesis

The audit trail below (Table 5) documents the key analytic decisions taken during the three-step synthesis. It provides traceability from coding to theme development (Themes T1-T4) and proposition development.

[Table 5 about here]

Table 5. Audit trail for the thematic synthesis leading to Themes T1–T4 and Propositions 1–4

Audit trail item	Decision/action documented
Synthesis step overview	Synthesis followed three steps: deductive coding aligned with RQ1-RQ3, inductive coding for emergent tensions and non-linear dynamics and cross-study comparison to derive integrative themes and propositions. Coding disagreements and theme boundaries were resolved via iterative codebook refinement, documented in this audit trail.
Deductive codebook basis	Initial deductive "code families" aligned with RQ1-RQ3: leadership behaviors and roles; mechanisms/mediators; contingencies/boundary conditions and governance routines; outcomes and levels of analysis. Definitions were stored in the codebook and applied consistently in the coding matrix columns.
Inductive codes added	During full-text reading, added inductive codes capturing emergent tensions (for example exploration-execution conflicts) and non-linear dynamics (for example overload). These notes were recorded in the "Tensions / non-linearities" column.
Codebook refinement rule	When overlap occurred, code definitions were refined and inclusion/exclusion rules added to keep leader behaviors (Theme T1) distinct from mechanisms (Theme T3), innovation strategy and portfolio logic (Theme T2) and dynamic capability and reconfiguration logic (Theme T4).
Theme clustering procedure	After coding all papers, compared recurring codes across studies and levels of analysis and clustered co-occurring concepts into candidate themes. Theme boundaries were iteratively refined until stable.
Theme derivation: T1	Clustered recurring leadership behavior codes (sensemaking and framing; enabling experimentation; orchestrating alignment) into Theme T1: Adaptive leadership behaviors in complex systems.

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Theme derivation: T2	Clustered innovation strategy codes (exploration and exploitation balance; sequencing; ambidexterity; innovation portfolio governance) into Theme T2: Innovation is portfolio-based and ambidextrous.
Theme derivation: T3	Clustered mechanism codes (psychological safety; trust; organizational support; self-efficacy and related mediators) into Theme T3: Mechanisms translating leadership into innovation.
Theme derivation: T4	Clustered reconfiguration codes (dynamic capabilities: sensing, seizing, transforming; scaling experiments; resource reallocation; governance routines) into Theme T4: Dynamic capabilities and reconfiguration.
Cross-cutting tensions handling	Tensions and failure modes were treated as cross-cutting (not a standalone theme) and retained in the "Tensions / non-linearities" column. They were used later when writing contingent statements and propositions.
Theme assignment rule	Each paper received one primary theme label (T1-T4) based on its main contribution. Cross-theme contributions were retained in the coded fields and in "Tensions / non-linearities" notes.
Evidence weighting decision	Quality appraisal signals were used to calibrate the strength of thematic statements and propositions (evidence weighting) rather than to exclude studies.

Appendix B. Concept-Centric Coding Matrix

To compare evidence across diverse study designs, we used a concept-centric coding matrix. In practical terms, we created a spreadsheet where each row represented an included article and each column captured a concept needed to answer RQ1–RQ3. This approach helps avoid treating heterogeneous studies as directly comparable to a single metric and instead synthesizes patterns at the level of concepts and mechanisms (Webster and Watson, 2002).

Core fields captured in the matrix were: (1) leadership construct and level of analysis; (2) focal leadership behaviors (e.g., sensemaking, enabling experimentation, alignment); (3) innovation construct and level (e.g., innovative work behavior, ambidexterity, innovation capability); (4) context/contingency descriptors (e.g., crisis intensity, technological turbulence, public accountability); (5) mechanisms/mediators (e.g., psychological safety, trust, support, efficacy); (6) boundary conditions and governance routines; and (7) the reported relationship or theoretical claim linking leadership to innovation.

Illustrative coded example: Diesel and Scheepers (2019) was coded as examining complexity leadership (leadership construct) at the level of organizational leadership (employee perceptions across hierarchical levels), with contextual ambidexterity operationalized as exploratory and exploitative innovation as the focal innovation construct, and organizational innovation climate as the mediating mechanism. This study therefore contributed evidence to

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Theme 2 (Portfolio-based innovation ambidexterity) and supported our mechanism emphasis in the propositions.

Appendix C. Included Studies and Extracted Evidence

Table 6 summarizes the 34 included studies and the extracted evidence used in the concept-centric synthesis. For each paper, we report its main contribution, method, assigned theme (T1–T4) and the appraisal tier used for evidence weighting (reported in Supplementary File S1). Together with Supplementary File S2, this overview supports traceability from primary studies to the thematic claims and propositions.

[Table 6 about here]

Table 6. Included studies in the integrative review (n=34): contribution, theme, method and appraisal tier

No.	Paper	Main contribution	Theme	Method	Quality tier	Notes
1	Ajith et al. (2025)	Examines operational-level adaptive crisis management responses to COVID-19 and how leaders adjust routines under turbulence.	T4	Case study / qualitative	High	Crisis management; resources sector; operational adaptation.
2	Anand et al. (2024)	Develops a pragmatist account of strategic renewal during crises and implications for renewal actions.	T4	Conceptual / theory-building	High	Crisis context; strategic renewal; theory building.
3	Bernards (2025)	Shows how cognitive uncertainty relates to employees' daily innovative work behavior and how ambidextrous leadership moderates this relationship.	T3	Quantitative study	High	Employee innovative behavior; uncertainty; ambidextrous leadership.
4	Biggadike et al. (2022)	Explores complexity leadership in project delivery contexts where time and budget constraints must be balanced with adaptation.	T1	Conceptual / theory-building	Moderate	Project/engineering management; balancing control and emergence.
5	Björck et al. (2025)	Develops a microfoundations perspective on cyber crisis readiness, highlighting leadership practices that support resilient action.	T4	Case study / qualitative	High	Cyber crisis context; readiness and resilience mechanisms.
6	Callens (2024)	Explores how complexity leadership can control and leverage network complexities to achieve collaborative innovation.	T1	Qualitative comparative analysis	High	Public administration; collaborative innovation; managing complexity.
7	Carroll et al. (2024)	Documents medical consultants' experience of collective leadership under complex conditions.	T1	Qualitative interview study	High	Healthcare; collective leadership; qualitative evidence.
8	Cousins et al. (2023)	Analyzes how hospitals build resilient information technology capabilities during crises across two contexts.	T4	Case study / qualitative	High	Health information technology; resilience; comparative crisis setting.

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9	Diesel and Scheepers (2019)	Empirically tests innovation climate as a mediator between complexity leadership and organizational ambidexterity.	T2	Quantitative study	Moderate	Innovation climate; complexity leadership; ambidexterity linkage.
10	El-Gazar et al. (2025)	Tests resilience as a mediator between ambidextrous leadership and nurses' positive attitudes toward artificial intelligence.	T3	Quantitative study	High	Healthcare; AI adoption attitudes; resilience as mechanism.
11	Fernández Fernández (2022)	Revisits March's "foolish" views to explain how leadership can cope with ambiguity and enable innovation.	T1	Conceptual / theory-building	Moderate	Ambiguity; leadership sensemaking; conceptual discussion.
12	Florek-Paszkowska and Ujwary-Gil (2025)	Proposes a digital-sustainability ecosystem framework linking digital transformation with sustainable innovation.	T4	Systematic review / conceptual framework	High	Digital transformation; sustainability; conceptual framework.
13	Gouda and Tiwari (2024)	Examines ambidextrous leadership as a pathway to build talent agility and engagement.	T2	Conceptual / viewpoint	Moderate	Human capital; agility and engagement outcomes supporting innovation.
14	Hannevig (2025)	Explores how chief executive officer (CEO) succession shapes ambidextrous strategies in family versus nonfamily small- and medium-sized enterprises (SMEs) through a knowledge- and resource-based view.	T2	Case study / qualitative	High	SMEs; leadership transition; ambidextrous strategy.
15	Ho et al. (2025)	Applies complexity leadership theory (with ecological leadership) to explain diffusion of an education innovation across schools.	T1	Case study / qualitative	High	Education innovation diffusion; cross-school leadership.
16	Indrianti et al. (2024)	Explores how entrepreneurial mindfulness and innovation capability contribute to entrepreneurial resilience within ambidextrous leadership.	T3	Quantitative study	Low	Entrepreneurship; resilience; mindfulness and innovation capability.
17	Jiang et al. (2023)	Provides a team science case study illustrating complexity leadership in action.	T1	Case study / qualitative	High	Research teams; coordination under complexity; case evidence.
18	Khan et al. (2022)	Tests how ambidextrous leadership relates to employee innovative performance through creative self-efficacy and self-resilience.	T3	Quantitative study	Low	Employee innovation; efficacy and resilience mechanisms.
19	Khan et al. (2025)	Examines academic and adaptive leadership during COVID-19 and its links to innovative behavior and organizational readiness for change.	T3	Quantitative study	High	Higher education; COVID-19 disruption; innovative behavior; organizational readiness for change; Pakistan.
20	Laufer et al. (2025)	Explores how university leaders foster innovation with educational technology through organizational trust.	T3	Qualitative interview study	High	Higher education; edtech innovation; trust-building.
21	Lestari et al. (2024)	Tests knowledge sharing as a mediator between clan culture, trust and resilience and innovative work behavior in Indonesian village-owned enterprises.	T3	Quantitative study	High	Village-owned enterprises; Indonesia; knowledge sharing; trust; resilience; innovative work behavior.

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22	Majumdarr et al. (2025)	Links digital transformational leadership and symmetrical internal communication with innovation capability through a moderated mediation model.	T3	Quantitative study	High	Digital leadership; internal communication; innovation capability.
23	Mendes et al. (2016)	Develops a conceptual framework linking complexity leadership roles (adaptive, enabling and administrative) to learning and innovation in organizations.	T1	Conceptual/theory -building	Moderate	Complexity leadership theory; learning; innovation; enabling conditions.
24	Nguyen-Van et al. (2025)	Examines green ambidextrous innovation and sustainable performance under VUCA conditions and the role of ambidextrous leadership.	T2	Quantitative study	High	Green innovation; VUCA; sustainable performance.
25	Oeij et al. (2016)	Examines how mindful infrastructure enables innovation resilience behavior in innovation teams via mechanisms such as psychological safety and team learning, alongside complexity leadership.	T3	Mixed	High	Innovation teams; mindful infrastructure; psychological safety; team learning; innovation resilience behavior.
26	Schindler et al. (2024)	Discusses how entrepreneurial culture and disruptive innovation interact with ambidexterity and proposes ways to handle tensions.	T2	Qualitative multiple-case study	High	Disruptive innovation; ambidexterity; organizational culture.
27	Schoemaker and Day (2021)	Outlines leadership imperatives for preparing organizations for greater turbulence.	T4	Conceptual / theory-building	Low	Turbulence; organizational preparedness; conceptual guidance.
28	Schophuizen et al. (2023)	Examines how complexity leadership enables educational innovation in university settings.	T1	Case study / qualitative	High	Higher education; complexity leadership; innovation implementation.
29	Schulze and Pinkow (2020)	Explains how enabling leaders create adaptive space to support organizational adaptability.	T1	Qualitative interview study	High	Adaptive space; enabling leadership; organizational adaptability.
30	Syamsir et al. (2025)	Systematically reviews leadership agility in a VUCA world and identifies conceptual insights and research directions.	T4	Systematic review	High	VUCA; leadership agility; synthesis of evidence.
31	Tsai et al. (2019)	Applies a complexity leadership lens to learning analytics adoption, identifying drivers, challenges and opportunities for innovation in socio-technical systems.	T1	Qualitative (interviews; thematic analysis)	High	Learning analytics; complexity leadership; digital/education innovation; socio-technical adoption challenges.
32	Uhl-Bien and Arena (2018)	Provides an integrative theoretical framework for leadership supporting organizational adaptability.	T4	Theoretical synthesis	High	Foundational post-2015 theory within sample; adaptability mechanisms.
33	Uhl-Bien (2021)	Conceptualizes how complexity leadership and followership are reshaped in a changed world.	T1	Conceptual / theory-building	Moderate	Complexity leadership; followership dynamics; conceptual.
34	Zuraik et al. (2020)	Examines gender differences in innovation and the role of team-lead ambidextrous leadership.	T2	Quantitative study	Moderate	Team-level leadership; innovation differences; ambidextrous leadership.

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